

# Random Hyperbolic Surface Spectral Rigidity — cycles 31-33

2026-05-16

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## Random Hyperbolic Surface Spectral Rigidity — cycles 31-33

### Abstract

Cycles 31-33 continued the local spectral-window branch opened in earlier work on Kim and Tao’s random-cover spectral rigidity paper. The previous cycle, M19, had closed the logarithmic-support smoothing route: a compactly supported trace test cannot resolve a polynomially shrinking spectral window unless its geometric support also grows polynomially. These three cycles asked what happens if that long support is allowed.

The answer was a staged narrowing of the remaining compact-support route. M20 quantified the variance saving needed when geometric support grows like a power of the cover degree. M21 converted that budget into a fixed-energy trace-side long-support variance theorem template. M22 then reduced the theorem template to a concrete localized Corollary 3.4 numerator target. No new local spectral statistics theorem was proved. The validated outcome is a precise next bottleneck: build a quotient-family model for the localized numerator  $p_{\Delta,q}$  before attempting a proof of long-support trace variance.

The final decision for this cycle range is therefore not closure of the local-window route, but further narrowing. The active branch is fixed-energy, trace-side, compact Paley-Wiener support with polynomial support parameter  $q = n^\eta$ . Pre-trace and edge-window branches remain out of scope because prior budgets made them less plausible.

### Introduction

The campaign studies the paper 2603.01127, “Eigenvalue rigidity of hyperbolic surfaces in the random cover model” by Elena Kim and Zhongkai Tao. The long-term goal is to understand the paper’s proof architecture and then search for credible extensions.

Earlier cycles established the following context.

M16 showed that subtracting Kim-Tao global Weyl-law estimates at two endpoints only controls spectral windows above an inherited global-error scale. This gives mesoscopic information, but not genuine shrinking local spectral statistics.

M17 formulated the missing kind of input: a smoothed-window variance estimate. For a window of width  $\Delta = n^{-d}$  in the bulk, a centered smoothed statistic has relative control if the variance exponent is small enough; in the notation used there,  $v/2 < 1 - d$ . To beat endpoint subtraction, the window exponent must also satisfy  $d > \alpha_W$ .

M18 mapped that target into the Kim-Tao test-function machinery and found a cost: inverse-width localization forces growth in the polynomial degree and geometric support parameter  $q$ , reintroducing the paper's  $q^{2\kappa}$  trace-side and  $q^{4\kappa}$  pre-trace losses.

M19 proved the smoothing obstruction that forced the next choice. For a compactly supported transform, resolving a bulk window  $\Delta = n^{-d}$  requires support  $R = n^\eta$  with  $\eta \geq d$ . Logarithmic support cannot resolve polynomially shrinking windows.

Cycles 31-33 start from that fork. The branch studied here accepts polynomial support and asks what new trace-side random-cover input would be needed.

## Approach

The work in cycles 31-33 followed a narrowing sequence.

1. M20 treated the missing random-cover saving as an unknown exponent  $\beta(\eta)$  and computed when that saving would overcome the long-support losses.
2. M21 converted the M20 budget into a theorem template, named the exact centered trace-side statistic, and identified which Kim-Tao inputs would need uniform long-support strengthening.
3. M22 moved one level closer to the proof object by defining the localized weighted numerator attached to Kim-Tao Lemma 3.3 and Corollary 3.4, then computed which coefficient-variation, direct small- $x$ , or stratified weighted bounds would imply the needed  $\beta$  saving.

This is a general mathematical report. It records validated work; it does not re-audit the conclusions.

## Source Inventory and Timeline

The following source sessions were gathered and used.

| Cycle | Session ID                           | Date       | Role       | Contents                                                                                        |
|-------|--------------------------------------|------------|------------|-------------------------------------------------------------------------------------------------|
| 31    | 169954ce-48cf-472a-a28b-a2db5ff61bbd | 2026-05-16 | researcher | Opened M20. Framed the long-support variance budget after M19 and specified required artifacts. |

| Cycle | Session ID                           | Date       | Role       | Contents                                                                              |
|-------|--------------------------------------|------------|------------|---------------------------------------------------------------------------------------|
| 31    | 25259870-a419-447a-a2ff-67239e24ee31 | 2026-05-16 | worker     | Built M20 proof note, report, analyzer, tests, CSVs, and figures.                     |
| 31    | 26b77f9e-e136-492b-9b7d-171698b029c6 | 2026-05-16 | auditor    | Validated M20 after one moderate repair to median computation in generated summaries. |
| 32    | 69efb200-e06c-4da7-935b-393262fdf225 | 2026-05-16 | researcher | Opened M21. Asked for the exact trace-side long-support theorem template.             |
| 32    | 7978fb53-3b20-4f6d-ab22-dd74fe347a34 | 2026-05-16 | worker     | Built M21 theorem-template proof note, report, analyzer, tests, CSVs, and figures.    |
| 32    | 97e2b0bd-de7c-4716-a0a7-5c4e9b562f15 | 2026-05-16 | auditor    | Validated M21 with no critical or moderate findings.                                  |
| 33    | 36c56594-f3e6-4cae-9d5e-430065bb65bc | 2026-05-16 | researcher | Opened M22. Narrowed the theorem target to the localized Corollary 3.4 numerator.     |
| 33    | ffacb468-1319-42d2-9539-d95f7c9cee99 | 2026-05-16 | worker     | Built M22 numerator target proof note, report, analyzer, tests, CSVs, and figures.    |
| 33    | 369b8204-7b26-4734-8b3a-e4afcb973eca | 2026-05-16 | auditor    | Validated M22 with no critical or moderate findings.                                  |

The main artifacts used were:

- docs/proof\_ledger/long\_support\_trace\_variance\_requirement.md

- reports/extension\_candidates/m20\_long\_support\_trace\_variance\_requirement.md
- docs/proof\_ledger/trace\_side\_long\_support\_variance\_template.md
- reports/extension\_candidates/m21\_trace\_side\_long\_support\_variance\_template.md
- docs/proof\_ledger/trace\_corollary34\_localized\_numerator\_target.md
- reports/extension\_candidates/m22\_trace\_corollary34\_uniform\_coefficient\_variation\_target.md
- the six generated CSV datasets under data/extension\_candidates/
- the six generated figures under reports/figures/

No REFERENCES.md file exists in the workspace, so this report cites internal source sessions and local artifacts rather than continuing a global numbered bibliography.

## Findings

### Finding 1: M20 Quantified the Long-Support Variance Budget

M20 asked what variance theorem would be required if the local-window trace route accepts polynomial geometric support.

The model starts with a future long-support variance estimate

$$\text{Var } Z_n \leq n^{1+L(\eta)-\beta(\eta)}.$$

Here  $L(\eta)$  records the support, degree, and interpolation loss, while  $\beta(\eta)$  is the new random-cover saving that is not available from existing Kim-Tao inputs.

For a bulk window  $\Delta = n^{-d}$ , the M17 smoothed-window Chebyshev condition becomes

$$\beta(\eta) > L(\eta) + 2d - 1.$$

At the edge, where the mean exponent is different, M20 recorded

$$\beta(\eta) > L(\eta) + 3d - 1.$$

M20 used the Kim-Tao support map from M18,

$$\text{supp}((h \circ f_{\Lambda_0})^\vee) \lesssim \Lambda_0^{-1/2} q,$$

so at fixed  $\Lambda_0$ , support  $R = n^\eta$  corresponds to  $q = n^\eta$ . The existing loss proxies are:

$$L_{\text{trace}}(\eta) = 2\kappa\eta, \quad L_{\text{pretrace}}(\eta) = 4\kappa\eta.$$

At minimal support and  $\kappa = 5$ , this gives the representative thresholds:

| Regime | Architecture | Required saving   |
|--------|--------------|-------------------|
| bulk   | trace        | $\beta > 12d - 1$ |
| bulk   | pre-trace    | $\beta > 22d - 1$ |
| edge   | trace        | $\beta > 8d - 1$  |
| edge   | pre-trace    | $\beta > 13d - 1$ |

The generated M20 budget table has 4,356 data rows. Its classes were:

| Class                        | Rows  |
|------------------------------|-------|
| impossible_by_support        | 1,872 |
| outside_current_architecture | 1,284 |
| requires_no_extra_saving     | 510   |
| requires_moderate_new_saving | 192   |
| requires_large_new_saving    | 498   |

The M20 auditor found one moderate issue: the summary script computed medians for even-cardinality lists using the upper-middle element rather than the true median. The auditor repaired the script, added a regression test, regenerated the data and figures, and confirmed that the formulas and strategic conclusion were unchanged.

For  $\kappa = 5$ , fixed  $\Lambda_0 = 4$ , and endpoint-beating rows satisfying the M19 support condition, the corrected medians were:

| Regime      | Architecture | Rows | Minimum beta | Median beta |
|-------------|--------------|------|--------------|-------------|
| bulk        | trace        | 28   | -0.904       | 0.15        |
| bulk        | pre-trace    | 28   | -0.824       | 1.15        |
| edge        | trace        | 44   | -0.942       | -0.091      |
| edge        | pre-trace    | 44   | -0.902       | 0.659       |
| high energy | trace        | 28   | -0.904       | 0.15        |
| high energy | pre-trace    | 28   | -0.824       | 1.15        |

The decision was to continue with a concrete long-support trace theorem template. The trace side remained conditionally plausible in a narrow exponent band; the pre-trace side was deprioritized because of the larger  $4\kappa\eta$  loss.

## Finding 2: M21 Turned the Budget into a Trace-Side Theorem Template

M21 named the exact fixed-energy trace-side statistic that the missing theorem should control.

Fix

$$\Lambda_0 > 1/4, \quad r_0 = \sqrt{\Lambda_0 - 1/4}, \quad \Delta = n^{-d}, \quad q = n^\eta, \quad \eta \geq d.$$

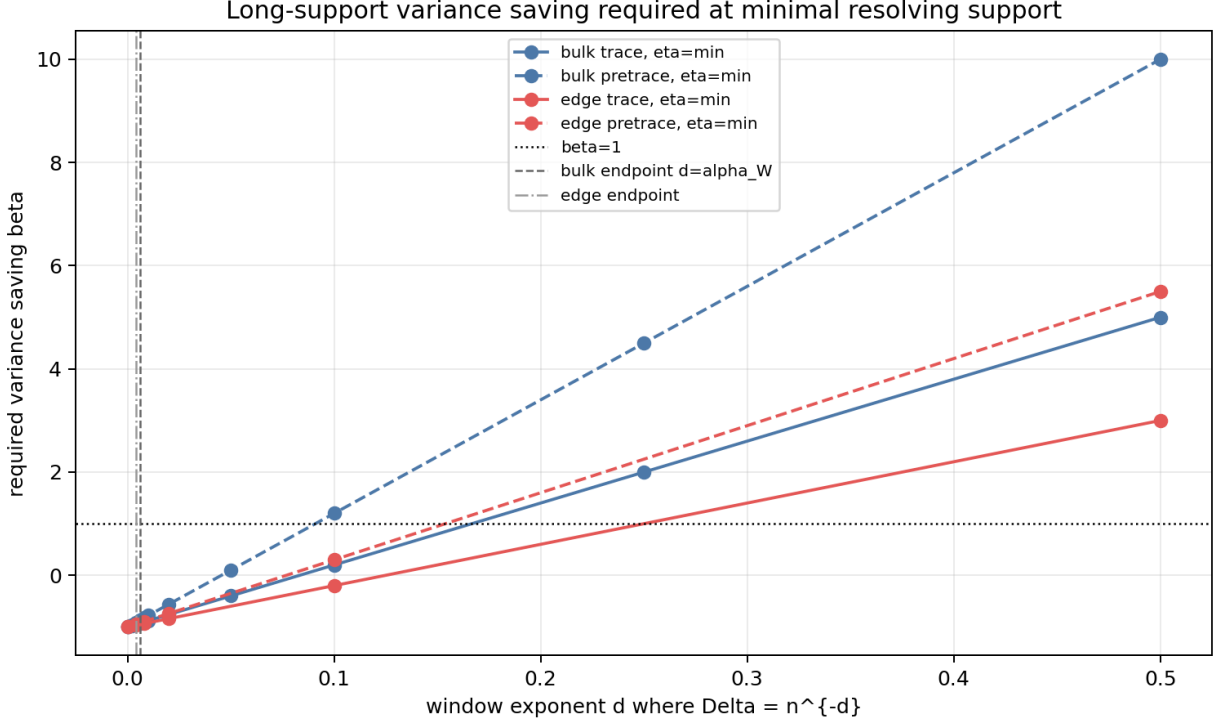
The localized spectral statistic is

$$Z_n(h_{\Lambda_0, \Delta, q}) = \sum_j h_{\Lambda_0, \Delta, q}(r_j(X_n)) - \frac{n \text{Vol}(X)}{2\pi} \int_0^\infty h_{\Lambda_0, \Delta, q}(r) r \tanh(\pi r) dr.$$

Here  $\lambda_j(X_n) = r_j(X_n)^2 + 1/4$ , and the  $r$ -window width is

$$\delta_r = \frac{\Delta}{2r_0} + O(\Delta^2).$$

By Kim-Tao Lemma 2.1, M21 identified the same statistic with the non-identity geometric trace sum



**Figure 1:** Required random-cover variance saving beta as a function of window exponent under trace and pre-trace long-support losses.

$$\sum_{\gamma \in P(X)} \sum_{k \geq 1} \frac{\ell_{\gamma}(X)}{2 \sinh(k \ell_{\gamma}(X)/2)} h_{\Lambda_0, \Delta, q}^{\vee}(k \ell_{\gamma}(X)) \operatorname{tr} \rho(\gamma^k).$$

This was the exact attachment point to the trace formula.

M21 then stated the conditional theorem template `LSTV_trace(eta,beta):`

$$\operatorname{Var} Z_n(h_{\Lambda_0, \Delta, q}) \leq C n^{1+2\kappa\eta-\beta+\epsilon}.$$

The term  $2\kappa\eta$  is the trace-side Markov/interpolation loss after substituting  $q = n^{\eta}$ . The new exponent  $\beta$  is the missing saving.

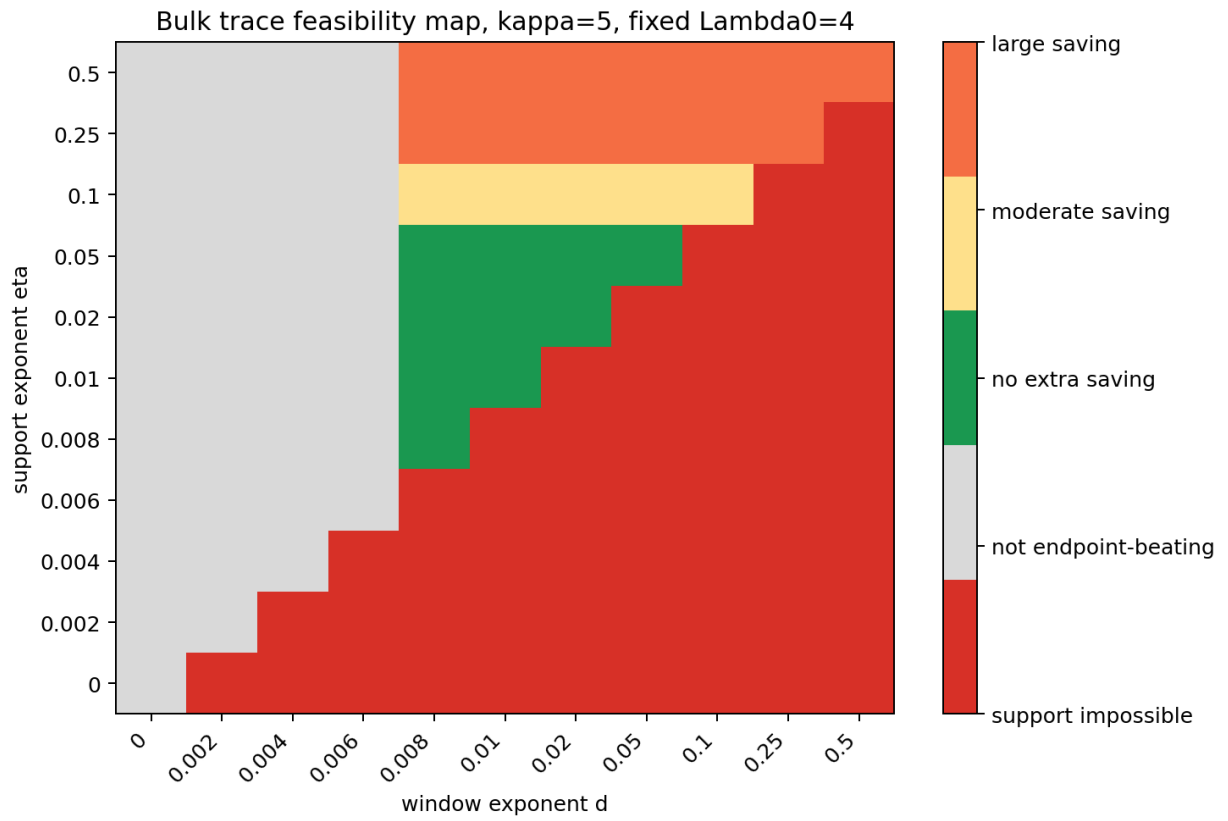
Since the bulk expected mass is  $\mu_n \asymp n\Delta = n^{1-d}$ , Chebyshev gives relative smoothed-window control when

$$1 + 2\kappa\eta - \beta < 2 - 2d,$$

or equivalently

$$\beta > 2\kappa\eta + 2d - 1.$$

To beat M16 endpoint subtraction, this must hold with  $d > \alpha_W$ .



**Figure 2:** Feasible, conditional, and obstructed regions for long-support local-window variance estimates.

The M21 analyzer produced 11,664 data rows and 81 summary rows. The generated classes were:

| Class                 | Rows  |
|-----------------------|-------|
| support_invalid       | 5,346 |
| not_endpoint_beating  | 3,564 |
| conditional_success   | 1,927 |
| needs_large_saving    | 531   |
| needs_moderate_saving | 296   |

For  $\kappa = 5$  and  $\alpha_W = 0.006$ , each beta model had 36 endpoint-beating, support-valid pairs  $(d, \eta)$ . The required beta range was

$$-0.904 \leq \beta_{\text{required}} \leq 2.0,$$

with median 0.07.

M21 did not claim the theorem is proved. Its dependency checklist separated existing inputs from missing ones:

| Dependency                                 | M21 status              |
|--------------------------------------------|-------------------------|
| Lemma 2.1 trace formula                    | existing                |
| M19 support condition                      | existing                |
| Markov interpolation framework             | existing but costly     |
| localized §2.4-style test construction     | new construction needed |
| Proposition 3.1 long-support variance form | needs uniform extension |
| Lemma 3.3 two-trace polynomial expansion   | needs uniform extension |
| Corollary 3.4 polynomial numerator         | new theorem-level input |
| MPvH/Witten-zeta normalization             | needs uniform extension |
| Nau boundedness                            | needs uniform extension |
| de-smoothing                               | out of scope            |

The M21 decision was: needs external uniform input before proof attempt.

### Finding 3: M22 Reduced the Theorem to the Localized Corollary 3.4 Numerator

M22 moved from the global theorem template to the upstream polynomial numerator target.

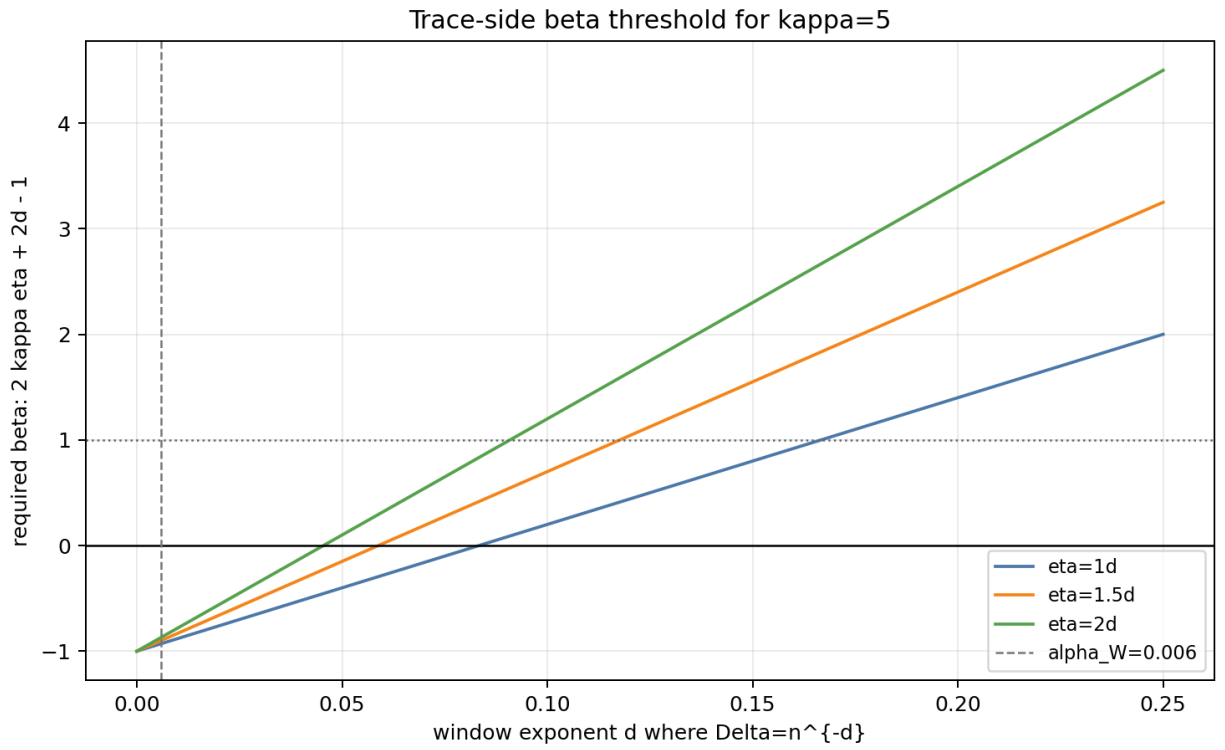
For a localized trace test  $h_{\Delta,q}$  with  $\Delta = n^{-d}$ ,  $q = n^\eta$ , and  $\eta \geq d$ , M22 defined

$$p_{\Delta,q}(x) = \sum_{\gamma_1, \gamma_2} \sum_{k_1, k_2 \geq 1} a(\gamma_1, k_1) a(\gamma_2, k_2) h_{\Delta,q}^\vee(k_1 \ell_{\gamma_1}) h_{\Delta,q}^\vee(k_2 \ell_{\gamma_2}) Q_{\gamma_1^{k_1}, \gamma_2^{k_2}}(x).$$

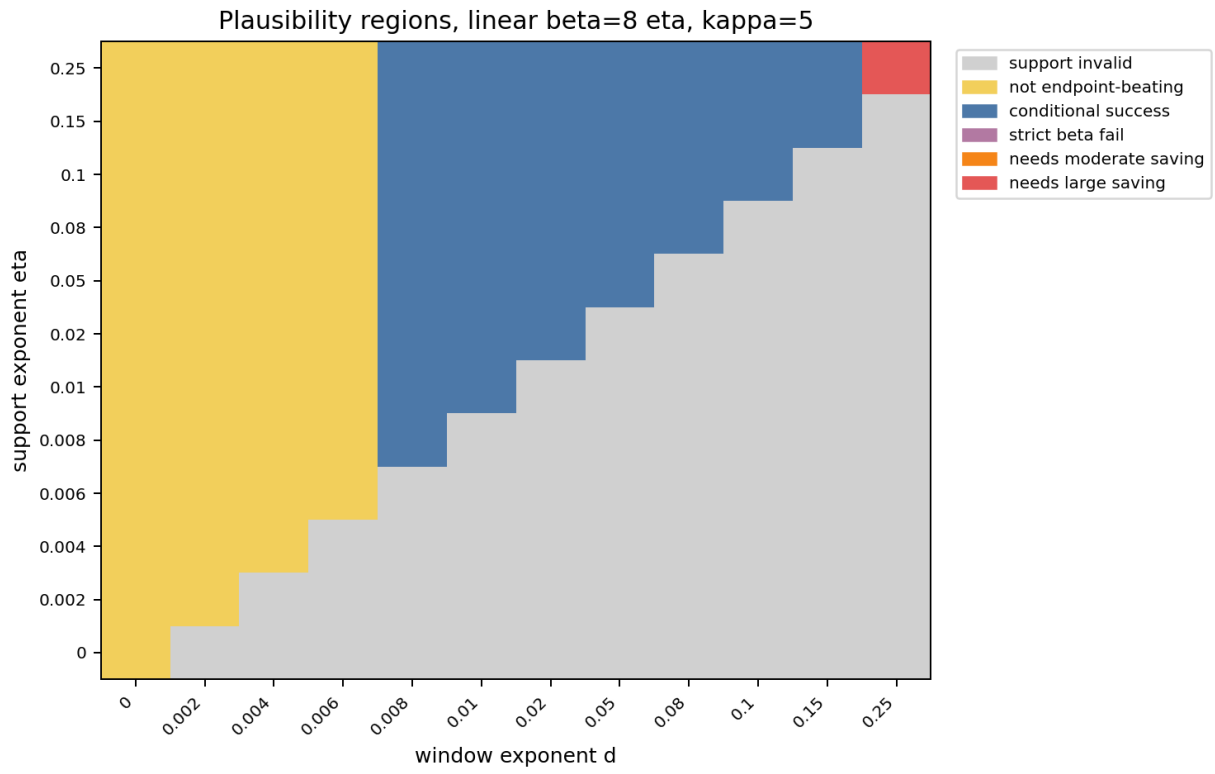
This is literally Kim-Tao’s Corollary 3.4 numerator only if the localized window can be realized inside the paper-compatible  $h \circ f_{\Lambda_0}$  polynomial/support architecture. Otherwise it is the nearest localized analogue and requires a new uniform Lemma 3.3 / Corollary 3.4 package.

The M22 beta algebra was:





**Figure 3:** Required trace-side long-support variance saving as a function of window exponent and support exponent.



**Figure 4:** Conditional success regions for candidate beta models in the fixed-energy trace-side theorem template.

$$\mathbb{E}G_n(h_{\Delta,q})^2 \leq nq^A n^{-\sigma+o(1)} \implies \beta = (2\kappa - A)\eta + \sigma.$$

This is measured relative to the M21 baseline  $nq^{2\kappa} = n^{1+2\kappa\eta}$ .

The fixed-bulk local-window condition stayed the same:

$$d > \alpha_W, \quad \eta \geq d, \quad \beta > 2\kappa\eta + 2d - 1.$$

M22 separated three sufficient target types:

| Target                | Meaning                                                               | Main risk                                      |
|-----------------------|-----------------------------------------------------------------------|------------------------------------------------|
| coefficient variation | absolute coefficient/tail control of $p_{\Delta,q}$ at $x = 1/n$      | may discard cancellation                       |
| direct small- $x$     | signed control of $p_{\Delta,q}(1/n)$ near $1/n$                      | cancellation only at $x = 0$ is not enough     |
| stratified weighted   | fixed $C - V$ or quotient-power strata with Selberg/localized weights | needs the actual quotient-family decomposition |

The pure Markov/interpolation baseline has  $A = 2\kappa$  and  $\sigma = 0$ , so  $\beta = 0$ . M22 found that this baseline succeeds only in rows where the required beta is already negative; it fails every endpoint-beating, support-valid row with positive required beta.

The M22 analyzer generated 10,368 data rows and 72 summary rows. The classes were:

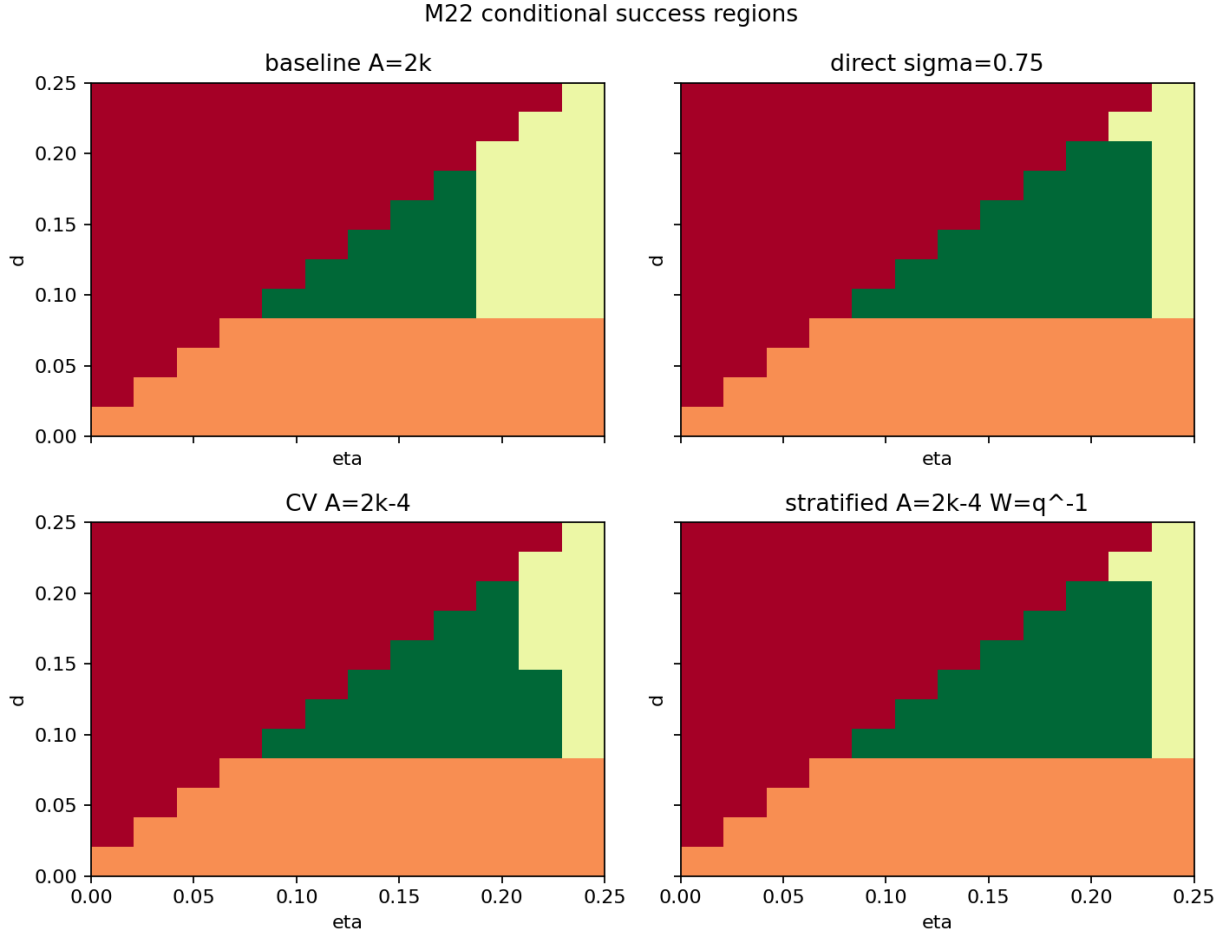
| Class                       | Rows  |
|-----------------------------|-------|
| support_invalid             | 4,752 |
| not_endpoint_beating        | 3,168 |
| conditional_success         | 1,498 |
| needs_more_numerator_saving | 950   |

For the representative  $\kappa = 5$ ,  $\alpha_W = 0.006$  endpoint-beating, support-valid band, selected target outcomes were:

| Target                              | Successes / 36 | Reading                                                 |
|-------------------------------------|----------------|---------------------------------------------------------|
| baseline A=2k                       | 15             | passes only negative-threshold rows                     |
| direct sigma=0.75                   | 27             | fixed small- $x$ saving handles most small-window rows  |
| CV A=2k-4                           | 24             | coefficient-variation improvement helps                 |
| stratified A=2k-4 W=q <sup>-1</sup> | 27             | stratified decay matches the tested direct-saving count |

The M22 decision was: needs quotient-family model before proof attempt.

This means the next cycle should model the actual weighted quotient-family strata appearing in  $p_{\Delta,q}$ , including Selberg weights, localized transform weights, and the  $Q_{\gamma_1^{k_1}, \gamma_2^{k_2}}$  numerator contribution.

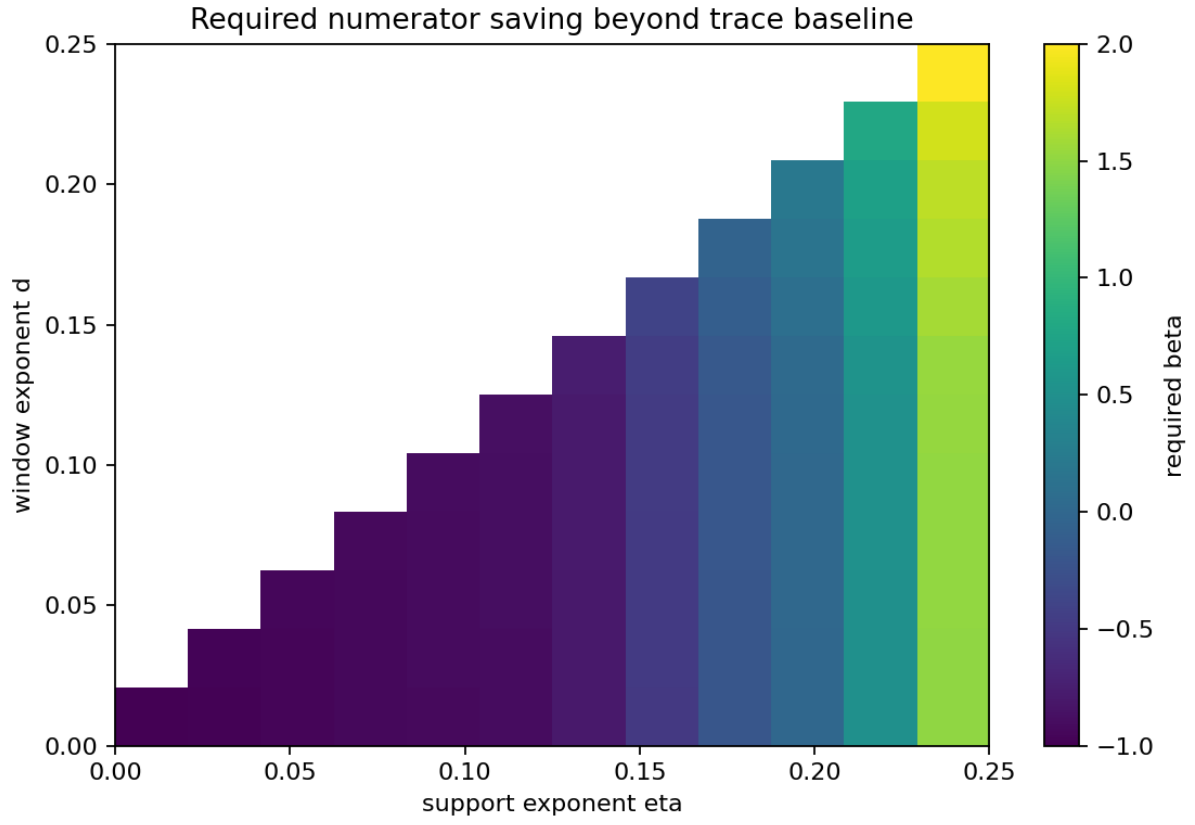


**Figure 5:** Conditional success regions for candidate localized numerator bounds in the fixed-energy trace-side bulk band.

#### **Finding 4: The Local-Window Route Has Been Narrowed to a Quotient-Family Numerator Model**

The cumulative logic through cycles 31-33 is:

1. M19 forced polynomial support for polynomially shrinking compact-support windows.
2. M20 showed that accepting polynomial support leaves a narrow trace-side exponent band, but requires a new variance saving  $\beta$ .
3. M21 stated the exact fixed-energy trace-side theorem template and found that the missing input is not simply Proposition 3.1 as a black box. It is uniform localized control around Lemma 3.3 and Corollary 3.4.
4. M22 defined the localized numerator target and showed which coefficient-variation, direct small- $x$ , or stratified weighted controls would produce useful beta saving.



**Figure 6:** Required numerator-level saving beyond the trace-side  $q^{2\kappa}$  loss as a function of  $d$  and  $\eta$ .

The live branch is therefore precise. It asks whether the localized numerator  $p_{\Delta,q}$  has enough quotient-family structure, cancellation, or weighted decay to produce positive  $\beta$  in the endpoint-beating, support-valid band.

The branch is still conditional. Existing Kim-Tao inputs do not prove this numerator saving, and the prior independent-permutation toy models do not automatically transfer to the surface-group trace numerator.

## Discussion

Cycles 31-33 did not produce a new rigidity theorem, but they made the local-window extension problem more inspectable.

Before M20, the phrase “long-support trace variance theorem” was a global slogan. After M21, it became a theorem template with a named statistic and a concrete inequality,

$$\text{Var } Z_n \leq n^{1+2\kappa\eta-\beta+o(1)}.$$

After M22, the theorem template became a numerator-level question about  $p_{\Delta,q}(x)$  at  $x = 1/n$ .

This matters because it separates three different problems that should not be conflated:

- constructing a localized test function compatible with Kim-Tao’s trace architecture;
- making Lemma 3.3 / Corollary 3.4 uniform when  $q = n^\eta$ ;
- proving actual numerator saving by coefficient variation, signed small- $x$  cancellation, or stratified weighted decay.

The validated guidance from the M22 audit is to build the quotient-family model next. A proof attempt before that model would lack the exact weighted strata it needs to control.

## Open Questions

1. Can the localized test  $h_{\Delta,q}$  be realized inside the  $h \circ f_{\Lambda_0}$  polynomial/support architecture without changing the proof object?
2. What are the actual quotient-family strata in the localized numerator  $p_{\Delta,q}$  once Selberg weights and localized transform weights are included?
3. Does  $p_{\Delta,q}$  exhibit coefficient variation, direct small- $x$  cancellation, or stratified weighted decay strong enough to give positive  $\beta$ ?
4. Can the MPvH/Witten-zeta, Nau boundedness, and MP23 rank-two decay inputs be made uniform for  $q = n^\eta$  in the endpoint-beating band?
5. If the quotient-family model fails to produce numerator saving, should the compact-support local-window route be closed in favor of noncompact geometric-tail trace methods?

## References

No REFERENCES.md file exists in the workspace. This report therefore cites the local source materials used for cycles 31-33:

- Kim and Tao paper files: 2603.01127.pdf, 2603.01127.txt.

- Cycle 31 sessions: 169954ce-48cf-472a-a28b-a2db5ff61bbd, 25259870-a419-447a-a2ff-67239e24ee31, 26b77f9e-e136-492b-9b7d-171698b029c6.
- Cycle 32 sessions: 69efb200-e06c-4da7-935b-393262fdf225, 7978fb53-3b20-4f6d-ab22-dd74fe347a34, 97e2b0bd-de7c-4716-a0a7-5c4e9b562f15.
- Cycle 33 sessions: 36c56594-f3e6-4cae-9d5e-430065bb65bc, ffacb468-1319-42d2-9539-d95f7c9cee99, 369b8204-7b26-4734-8b3a-e4afcb973eca.
- M20 artifacts: docs/proof\_ledger/long\_support\_trace\_variance\_requirement.md, reports/extension\_candidates/m20\_long\_support\_trace\_variance\_requirement.md, data/extension\_candidates/long\_support\_variance\_budget.csv, data/extension\_candidates/long\_support\_variance\_summary.csv.
- M21 artifacts: docs/proof\_ledger/trace\_side\_long\_support\_variance\_template.md, reports/extension\_candidates/m21\_trace\_side\_long\_support\_variance\_template.md, data/extension\_candidates/trace\_variance\_template\_budget.csv, data/extension\_candidates/trace\_variance\_template\_summary.csv.
- M22 artifacts: docs/proof\_ledger/trace\_corollary34\_localized\_numerator\_target.md, reports/extension\_candidates/m22\_trace\_corollary34\_uniform\_coefficient\_variation\_target.md, data/extension\_candidates/corollary34\_target\_budget.csv, data/extension\_candidates/corollary34\_target\_summary.csv.

## Appendix: Implementation Details

### Code Organization

Cycle 31 added:

| File                                                                        | Lines | Purpose                                                                                                         |
|-----------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------|
| scripts/analyze_long_support_variance_budget.py                             | 278   | Computes M20 long-support variance-saving budgets for trace and pre-trace local-window routes.                  |
| tests/test_long_support_variance_budget.py                                  | 99    | Tests M20 support thresholds, loss ratios, Chebyshev inequalities, endpoint logic, and true median computation. |
| docs/proof_ledger/long_support_trace_variance_requirement.md                | 180   | Records the M20 exponent-budget theorem template.                                                               |
| reports/extension_candidates/m20_long_support_trace_variance_requirement.md | 113   | Summarizes the M20 decision and generated diagnostics.                                                          |

Cycle 32 added:

| File                                                                          | Lines | Purpose                                                                                             |
|-------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------|
| scripts/analyze_trace_variance_template_budget.py                             | 268   | Computes M21 beta-model budgets for the fixed-energy trace-side theorem template.                   |
| tests/test_trace_variance_template_budget.py                                  | 94    | Tests M21 required beta, support validity, endpoint-beating logic, and no-saving baseline behavior. |
| docs/proof_ledger/trace_side_long_support_variance_template.md                | 191   | States the trace-side long-support theorem template and dependency map.                             |
| reports/extension_candidates/m21_trace_side_long_support_variance_template.md | 160   | Summarizes the M21 theorem target and decision.                                                     |

Cycle 33 added:

| File                                                                                       | Lines | Purpose                                                                                             |
|--------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------|
| scripts/analyze_corollary34_target_budget.py                                               | 245   | Computes M22 localized numerator target budgets and success regions.                                |
| tests/test_corollary34_target_budget.py                                                    | 90    | Tests M22 beta algebra, support validity, endpoint logic, monotonic savings, and baseline failures. |
| docs/proof_ledger/trace_corollary34_localized_numerator_target.md                          | 228   | Defines the localized numerator and sufficient target classes.                                      |
| reports/extension_candidates/m22_trace_corollary34_uniform_coefficient_variation_target.md | 111   | Summarizes the M22 numerator target and proof-facing decision.                                      |

## Generated Data

| Dataset                                                     | Data rows |
|-------------------------------------------------------------|-----------|
| data/extension_candidates/long_support_variance_budget.csv  | 4,356     |
| data/extension_candidates/long_support_variance_summary.csv | 12        |



| Dataset                                                       | Data rows |
|---------------------------------------------------------------|-----------|
| data/extension_candidates/trace_variance_template_budget.csv  | 11,664    |
| data/extension_candidates/trace_variance_template_summary.csv | 81        |
| data/extension_candidates/corollary34_target_budget.csv       | 10,368    |
| data/extension_candidates/corollary34_target_summary.csv      | 72        |

## Figure Inventory

| Figure                                                      | Size        |
|-------------------------------------------------------------|-------------|
| reports/figures/m20_required_variance_saving.png            | 1530 x 936  |
| reports/figures/m20_long_support_feasibility_map.png        | 1475 x 1007 |
| reports/figures/m21_trace_template_beta_thresholds.png      | 1494 x 936  |
| reports/figures/m21_trace_template_plausibility_regions.png | 1512 x 972  |
| reports/figures/m22_corollary34_target_success_regions.png  | 1440 x 1120 |
| reports/figures/m22_required_numerator_saving.png           | 1152 x 800  |

## Validation Results

Cycle validators reported the following:

- M20: validated after one audit repair to true median computation. Python compile, analyzer, tests, figure checks, promise\_check, and org\_check passed.
- M21: validated with no critical or moderate findings. Python compile, analyzer, tests, independent formula checks, figure checks, promise\_check, and org\_check passed.
- M22: validated with no critical or moderate findings. Python compile, analyzer, tests, independent CSV formula checks, figure checks, promise\_check, and org\_check passed.

After the reporter manifest update, the following checks were rerun:

```
python3 -m long_exposure.tools.promise_check .
events: 88, plan milestones: 22
exit code: 0, historical warnings only

python3 -m long_exposure.tools.org_check .
exit code: 0, historical warnings only
```

The historical promise\_check warnings concern one noncanonical old docs/paper\_map/ artifact path and orphaned prior periodic reports. The historical org\_check warnings concern root paper/live-run files and older figures under docs/. These warnings predate cycles 31-33.

## Manifest Snapshot

MANIFEST.md was replaced with a current snapshot after M22. It contains:

| Metric                                                                      | Value |
|-----------------------------------------------------------------------------|-------|
| Manifest lines                                                              | 154   |
| Campaign scripts                                                            | 35    |
| Campaign script lines                                                       | 9,000 |
| Campaign test files                                                         | 23    |
| Campaign test lines                                                         | 1,892 |
| Markdown/DOT/PNG documentation artifacts under docs/, reports/, and audits/ | 129   |
| PNG figures under reports/figures/                                          | 55    |
| Canonical CSV datasets under data/                                          | 60    |
| Promise ledger events                                                       | 88    |

There was no ## Key Files section in the prior manifest, so no final-reporter-owned section needed preservation.

## Cross-Reference Map

| Origin                  | Consuming artifact   | Role                                                                                           |
|-------------------------|----------------------|------------------------------------------------------------------------------------------------|
| M17 variance criterion  | M20 budget           | Supplies the Chebyshev variance-to-window-count inequality.                                    |
| M18 support map         | M20 budget           | Converts long support into $q$ growth and trace/pre-trace losses.                              |
| M19 Fourier obstruction | M20 budget           | Forces $\eta \geq d$ in the bulk and $\eta \geq d/2$ at the edge.                              |
| M20 long-support budget | M21 theorem template | Selects fixed-energy trace-side variance as the plausible compact-support continuation.        |
| M21 theorem template    | M22 numerator target | Narrows the missing variance theorem to localized Lemma 3.3 / Corollary 3.4 numerator control. |
| M22 numerator target    | next milestone       | Recommends M23-localized-trace-numerator-quotient-family-model.                                |